Lab Mid Exam: HOG vs SIFT on Blood-Cell Images

Computer Vision Lab
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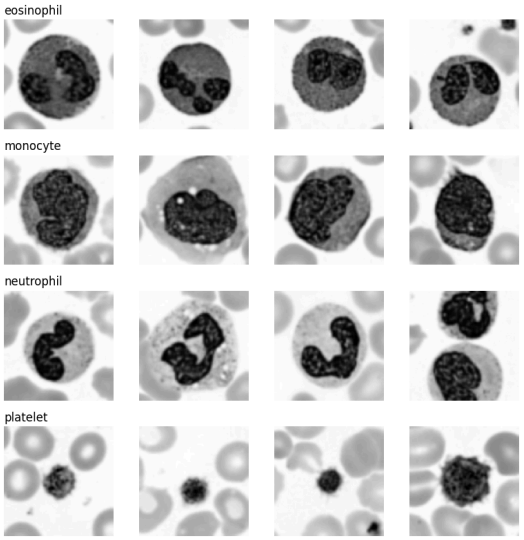
Personal Settings

Setting	Rule	My value
Random seed	Last two digits of 044	44
HOG cell size	Last digit 4 is even	8 x 8
SIFT grid size	Last digit 4 is in 0 to 4	4 x 4 grid (16 keypoints)

Q1. Data Setup

Classes eosinophil (1), monocyte (5), neutrophil (6) and platelet (7) were kept from BloodMNIST (size 64). Using seed 44, 100 train, 30 validation and 30 test images were picked per class, converted to grayscale and resized to 128 x 128.

```
Train (400, 128, 128) {'eosinophil': 100, 'monocyte': 100, 'neutrophil': 100,
'platelet': 100}
Val (120, 128, 128) {'eosinophil': 30, 'monocyte': 30, 'neutrophil': 30, 'platelet':
30}
Test (120, 128, 128) {'eosinophil': 30, 'monocyte': 30, 'neutrophil': 30, 'platelet':
30}
```



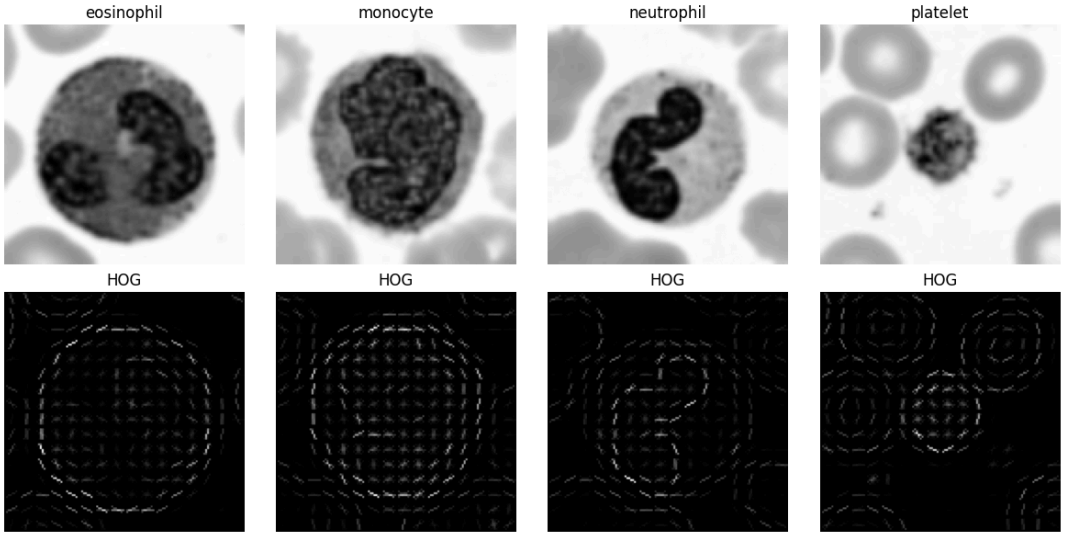
Q2. HOG + SVM

Feature length

```
HOG feature length: 8100
```

Step	Calculation
Cells per side	128 / 8 = 16
Blocks per side (2 x 2 cells, slides 1 cell)	16 - 2 + 1 = 15
Total blocks	15 x 15 = 225
Values per block	2 x 2 cells x 9 orientations = 36
Feature length	**225 x 36 = 8100**

HOG picture for one image per class

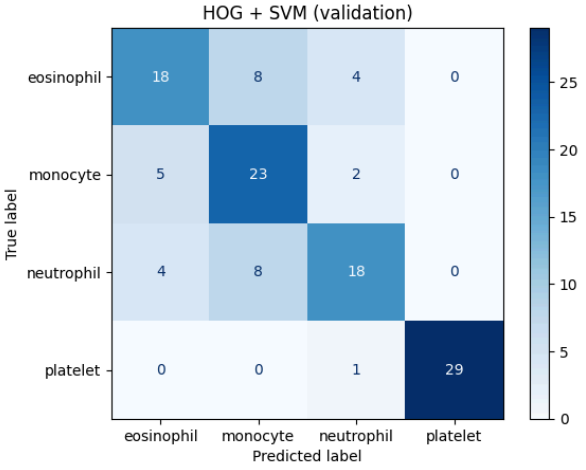


SVM result (C chosen on validation)

```
C = 0.1: validation accuracy = 0.733
C = 1: validation accuracy = 0.733
C = 10: validation accuracy = 0.733

Best C = 0.1
Validation accuracy = 0.733
Validation macro-F1 = 0.734
```

> **HOG + SVM:** best C = 0.1 (all three C values tied at 0.733, so the first was kept), validation accuracy = 0.733, validation macro-F1 = 0.734.



Q3. SIFT + SVM

Normal SIFT detection on training images

```
Average keypoints per image: 70.67
Minimum keypoints: 10
Maximum keypoints: 240
Images with zero keypoints: 0
```

Average keypoints	Minimum	Maximum	Images with zero keypoints
70.67	10	240	0

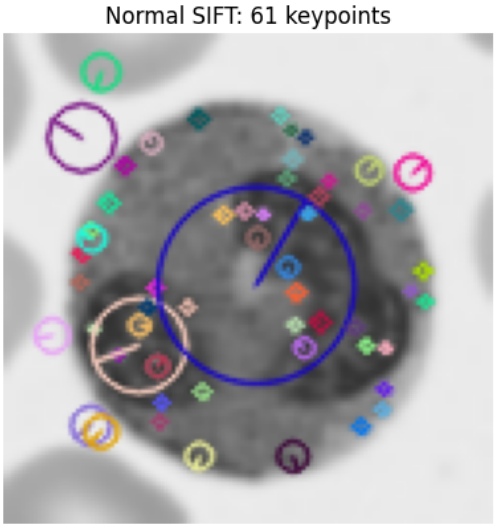
Why is this a problem for an SVM? An SVM needs every image to give a feature vector of the same length, but normal SIFT gives a different number of keypoints per image (from 10 to 240 here), so the descriptors cannot be stacked into one fixed size vector.

Fixed grid of keypoints

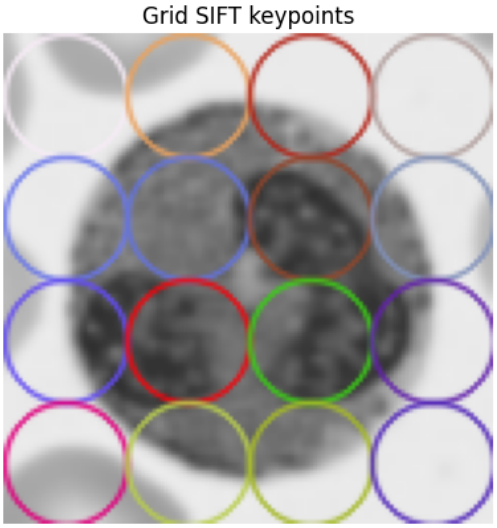
A 4 x 4 grid was placed over the 128 x 128 image with points 32 pixels apart (centres at 16, 48, 80, 112) and keypoint size 32. `sift.compute` gives a 16 x 128 array for each image, which is flattened into one vector.

```
Grid points: 16
SIFT feature length: 2048
```

Step	Calculation
Grid points	4 x 4 = 16
One SIFT descriptor	4 x 4 sub-regions x 8 orientation bins = 128
Feature length	**16 x 128 = 2048**



Normal SIFT (61 keypoints on this image)



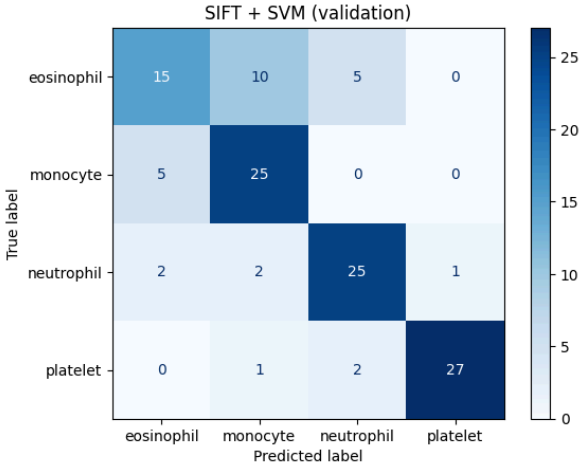
Fixed 4 x 4 grid (16 keypoints)

SVM result (C chosen on validation)

```
C = 0.1: validation accuracy = 0.758
C = 1: validation accuracy = 0.767
C = 10: validation accuracy = 0.767

Best C = 1
Validation accuracy = 0.767
Validation macro-F1 = 0.762
```

SIFT + SVM: best C = 1 (C = 1 and C = 10 tied at 0.767, so the first was kept), validation accuracy = 0.767, validation macro-F1 = 0.762.



Q4. Robustness

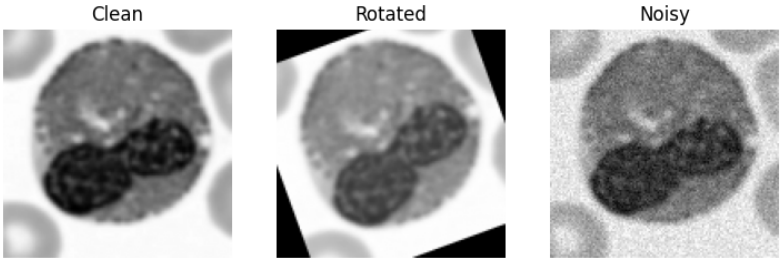
Prediction (written before running)

Rotation: HOG will lose more accuracy. HOG uses small 8 x 8 cells at fixed positions and fixed gradient directions, so when the image turns by 20 degrees the gradients move into different bins and cells.

Noise: HOG will lose more accuracy. Noise creates fake gradients, and HOG's small 8 x 8 cells feel this a lot, while each SIFT descriptor covers a bigger 32 pixel patch with Gaussian weighting so the noise gets averaged out more.

Test images after each change

Both models were frozen (only `predict` was called). Each change was applied to the test images before feature extraction: rotation by 20 degrees about the centre, and Gaussian noise with sigma 0.05 on the image scaled to 0 to 1.



Results

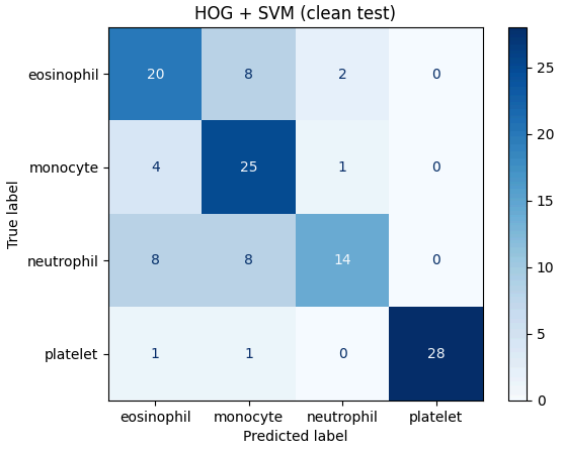
```
Model           Clean   Rotated     Noisy
HOG + SVM       0.725     0.658     0.250
SIFT + SVM      0.750     0.467     0.750
```

Model	Clean accuracy	Rotated accuracy	Noisy accuracy
HOG + SVM	0.725	0.658	0.250
SIFT + SVM	0.750	0.467	0.750

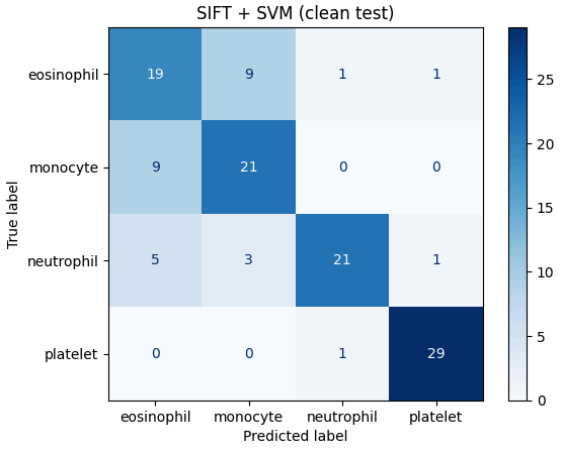
Was my prediction correct?

My noise prediction was correct: HOG fell from 0.725 to 0.250, which is chance level for four classes, while SIFT stayed at 0.750. My rotation prediction was wrong: HOG only dropped from 0.725 to 0.658 (6.7 points) but SIFT dropped from 0.750 to 0.467 (28.3 points). A likely reason is that my 16 grid keypoints have fixed positions and a fixed orientation, so after a 20 degree turn every large patch (including the four corner patches that now hold black fill) describes different content, while for noise SIFT computes each size 32 descriptor on a blurred level of the image, which smooths the noise away.

Confusion matrices on the clean test set



HOG + SVM (clean test)



SIFT + SVM (clean test)

Q5. Short Answers and AI Log

(a) Why does HOG use block normalisation?

Lighting and stain contrast change from image to image, so the same cell shape can give strong gradients in one image and weak ones in another. Block normalisation divides the histograms of each 2 x 2 block by their overall size, so the feature describes the shape of the edges and not how bright or dark the image is. Because blocks overlap, each cell gets normalised several times against different neighbours, which makes the feature more stable.

(b) Why is it wrong to choose C using the test set?

If we pick C by looking at test accuracy, the test set has been used to tune the model, so the reported test score is too optimistic. The test set has to stay unseen until the end so it gives an honest estimate of performance on new images, which is why C was chosen on the validation set and the test set was only used in Q4.

Which two classes did the models confuse most, and why might grayscale make this harder?

Both models confused eosinophil and monocyte the most. On the clean test set HOG predicted 8 eosinophils as monocytes and 4 monocytes as eosinophils, and SIFT made the same mistake 9 times in each direction (validation shows the same pair: 8 + 5 for HOG and 10 + 5 for SIFT). Grayscale makes this harder because the main clue between them is colour: eosinophils have bright orange red granules from the eosin stain while monocytes have pale blue grey cytoplasm. In grayscale both become a round grey cell with a large dark nucleus of similar size, so only the nucleus shape and texture is left to separate them.

AI log

Prompt (one line)	What I changed	How I checked it was correct
Understand the exam and write the simplest HOG + SVM and grid SIFT + SVM code for reg no 044	Set SEED = 44, HOG cell 8 x 8 and a 4 x 4 SIFT grid. Added StandardScaler before LinearSVC because SIFT values go up to about 150 while HOG values are small.	Printed shapes and class counts (400 / 120 / 120, exactly 100 / 30 / 30 per class). HOG length 8100 matched my hand calculation and SIFT length 2048 matched 16 x 128. Looked at the HOG pictures and the grid keypoint image.
Colab says No module named cv2	Added opencv-python-headless and scikit-image to the pip install cell, because the TPU runtime does not include OpenCV.	Ran all cells again. Every import worked and every cell produced output.